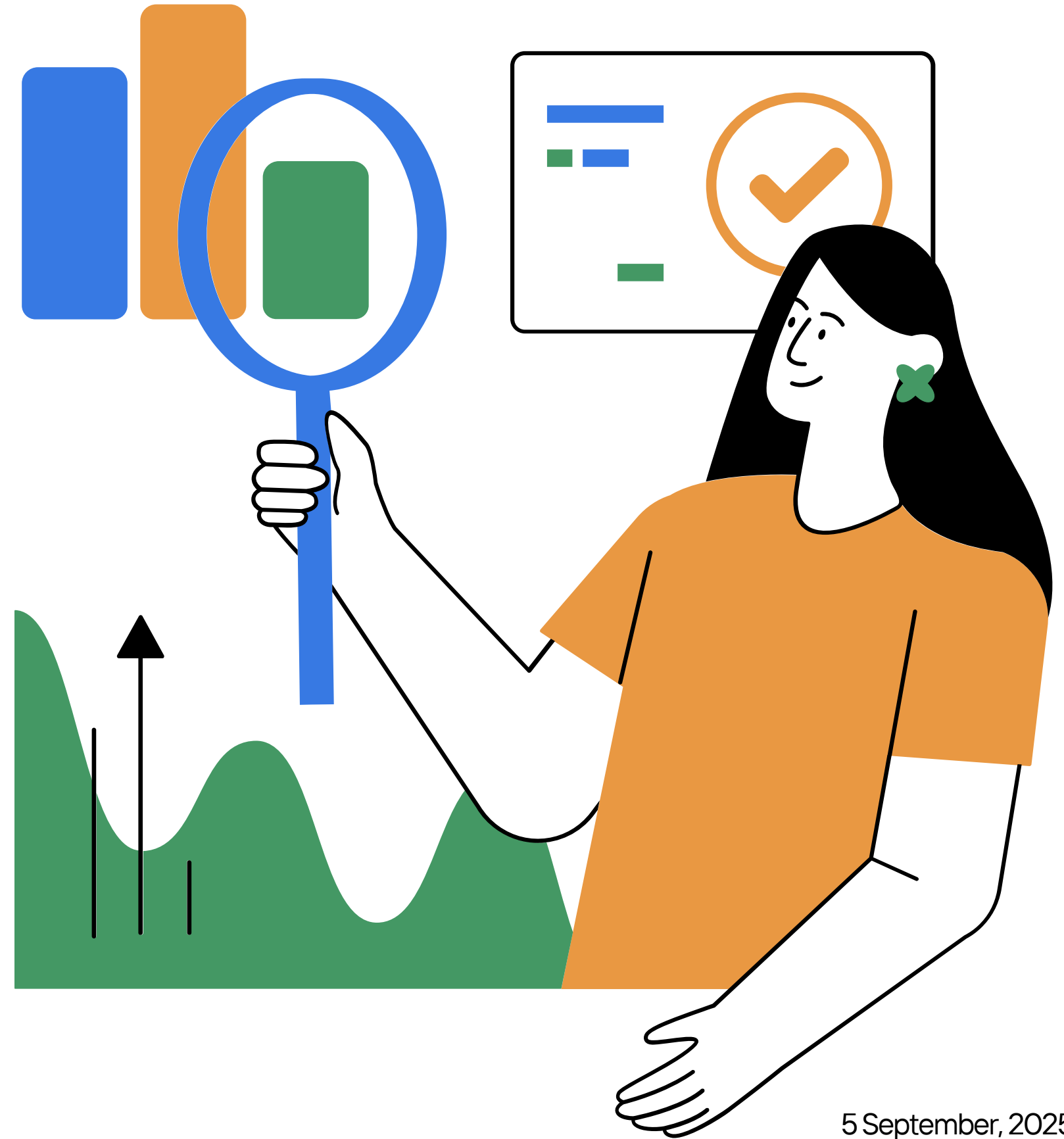


# Employee Attrition Analysis & Prediction



5 September, 2025

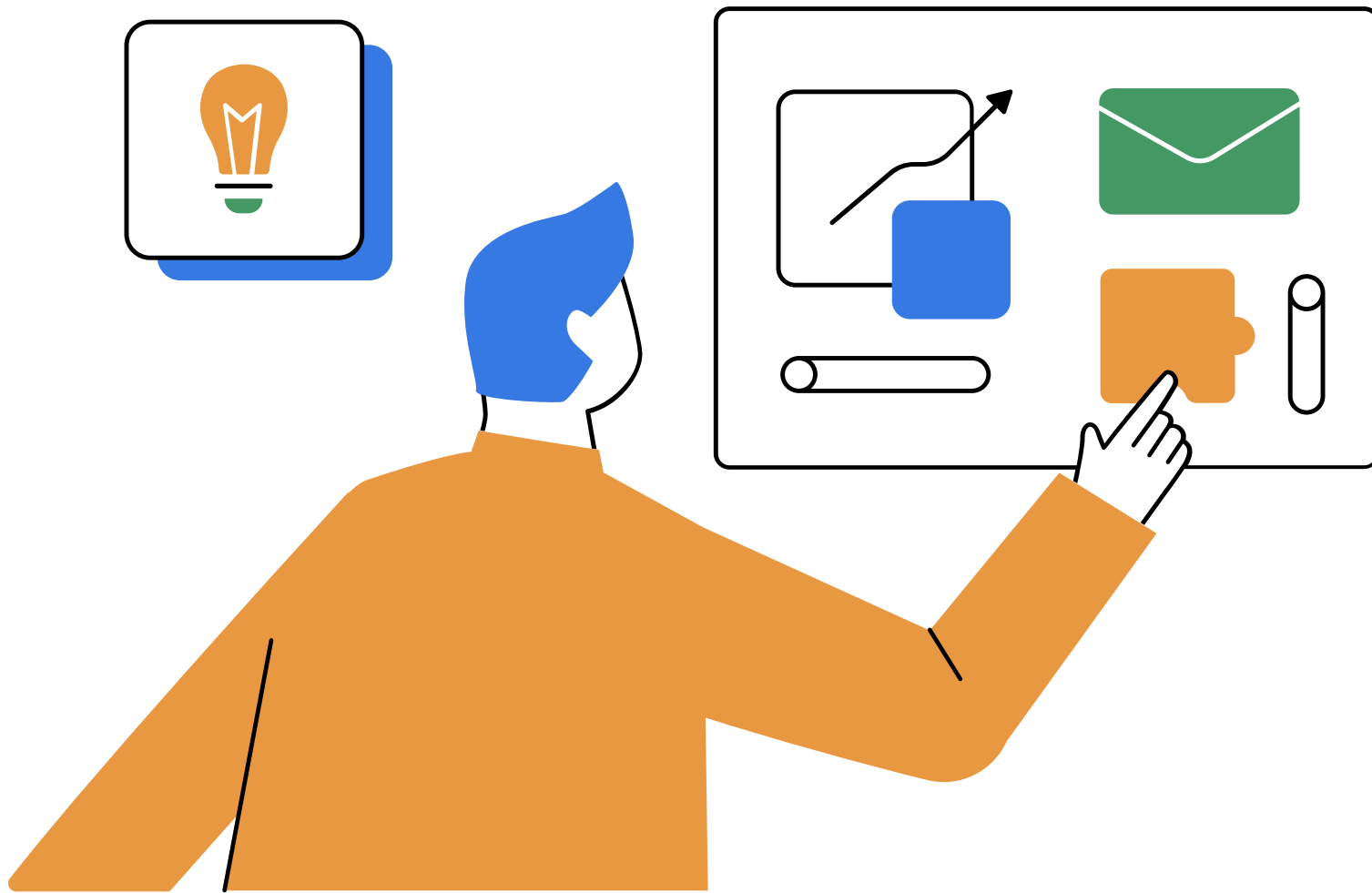


# Team Members

- Van Kirk Lumantas
- Kent Spencer Medez
- Bona Ruth Barol
- Jean Mar Tuanda

# Agenda

- Problem Overview
- Data Overview
- Preprocessing
- Exploratory Data Analysis
- Data Analysis
- Report and Recommendation



# Objectives

- **Identify the main reasons why employees leave** –  
Understanding root causes of turnover
- **Find out which departments have the highest turnover** –  
Pinpointing problem areas within the organization
- **Discover patterns** – Analyzing demographic and professional factors
- **Build a model that can predict employee attrition** –  
Creating predictive tools for proactive management

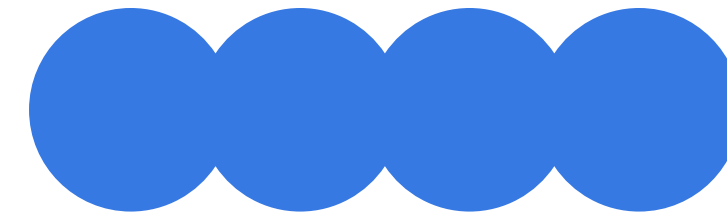


# The Problem

- Employee turnover creates major costs:
  - High recruitment and training expenses
  - Loss of experienced talent
  - Reduced productivity
  - Negative impact on company profits







# Dataset Overview

- Source: IBM HR Analytics Dataset from Kaggle
- Size: 1,470 employees, 35 features



Source:

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset/>

- Age
- Attrition
- BusinessTravel
- DailyRate
- Department
- DistanceFromHome
- Education
- EducationField
- EmployeeCount
- EmployeeNumber
- EnvironmentSatisfaction
- Gender
- HourlyRate
- JobInvolvement
- JobLevel

- JobRole
- JobSatisfaction
- MaritalStatus
- MonthlyIncome
- MonthlyRate
- NumCompaniesWorked
- Over18
- OverTime
- PercentSalaryHike
- PerformanceRating
- RelationshipSatisfaction
- StandardHours
- StockOptionLevel
- TotalWorkingYears
- TrainingTimesLastYear

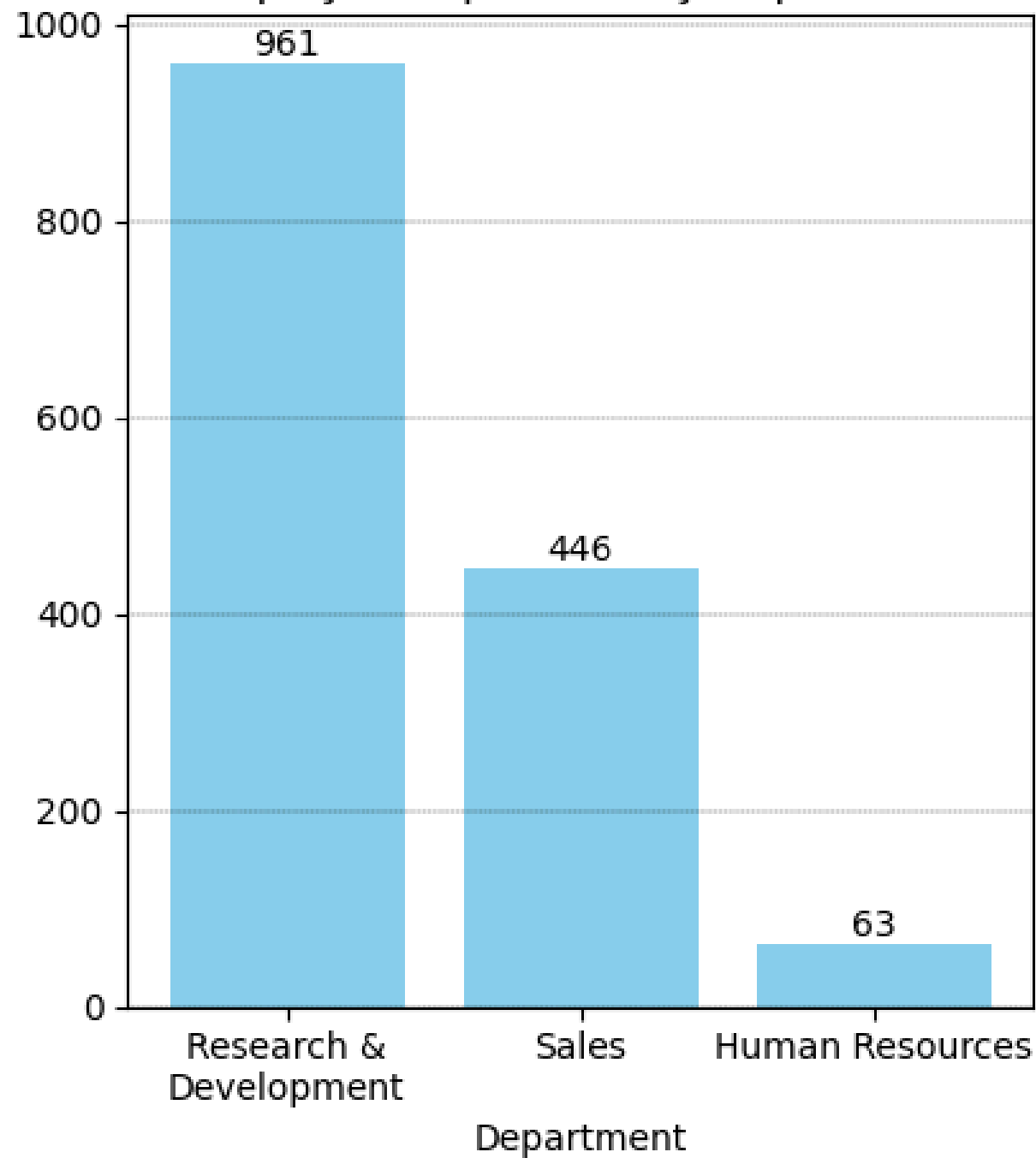
- WorkLifeBalance
- YearsAtCompany
- YearsInCurrentRole
- YearsSinceLastPromotion
- YearsWithCurrManager



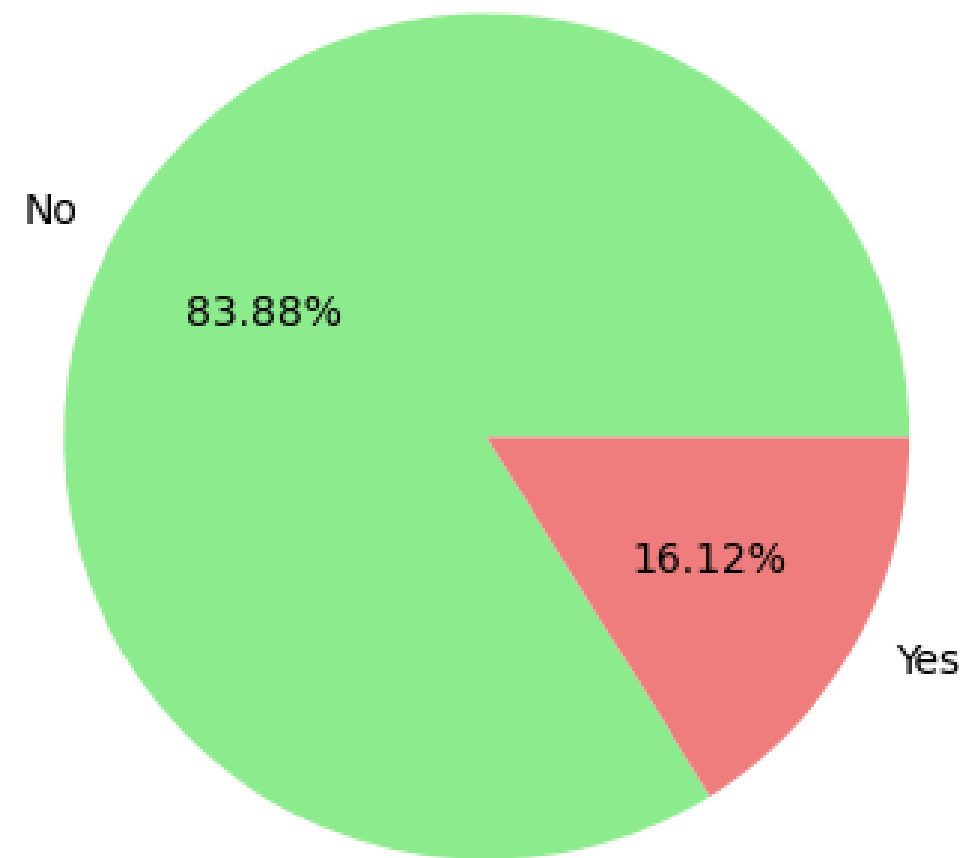
**Attrition is at 16.12%  
for 1,470 employees.**



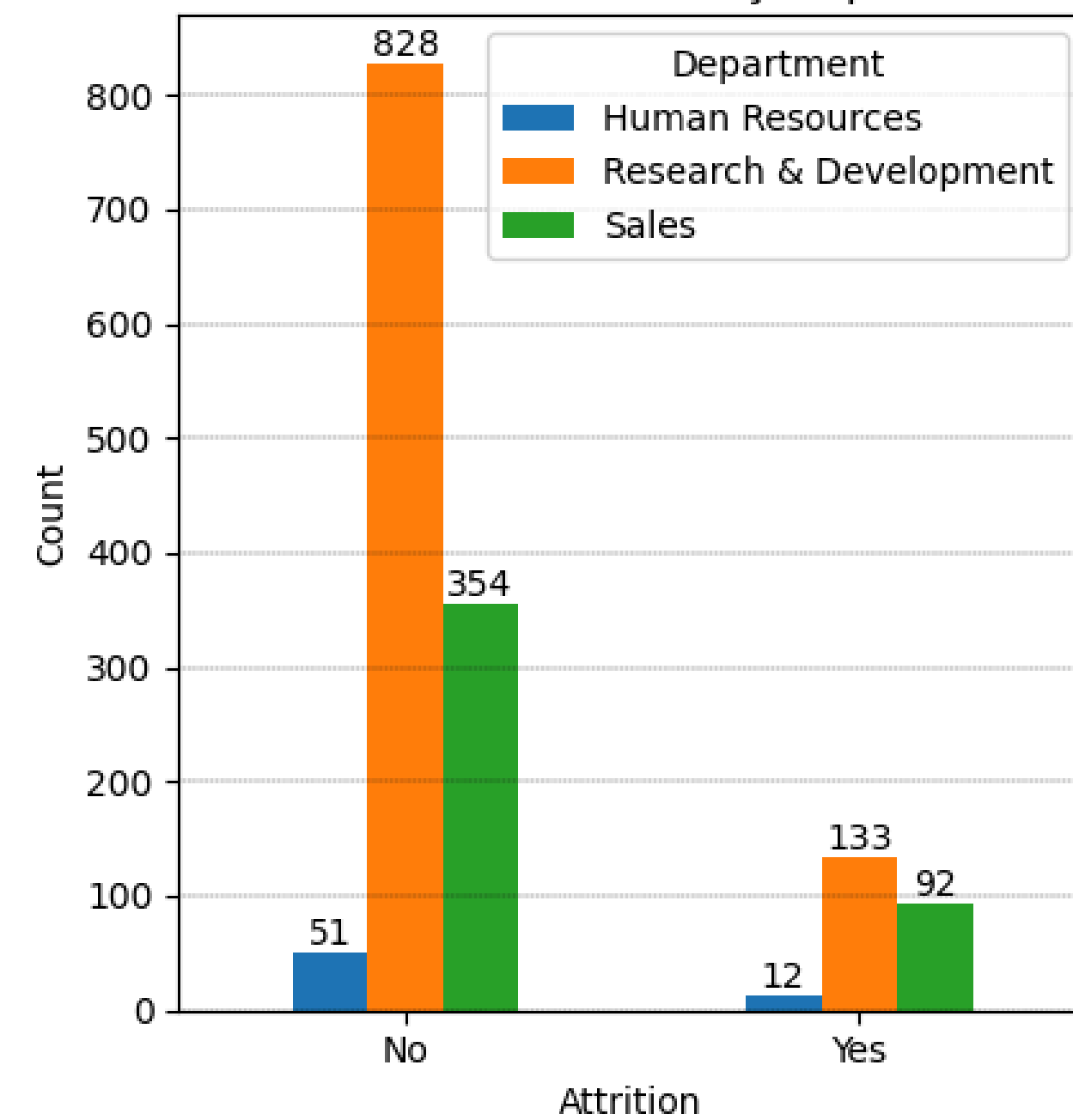
Employee Population By Department



Attrition and Retention Percentage



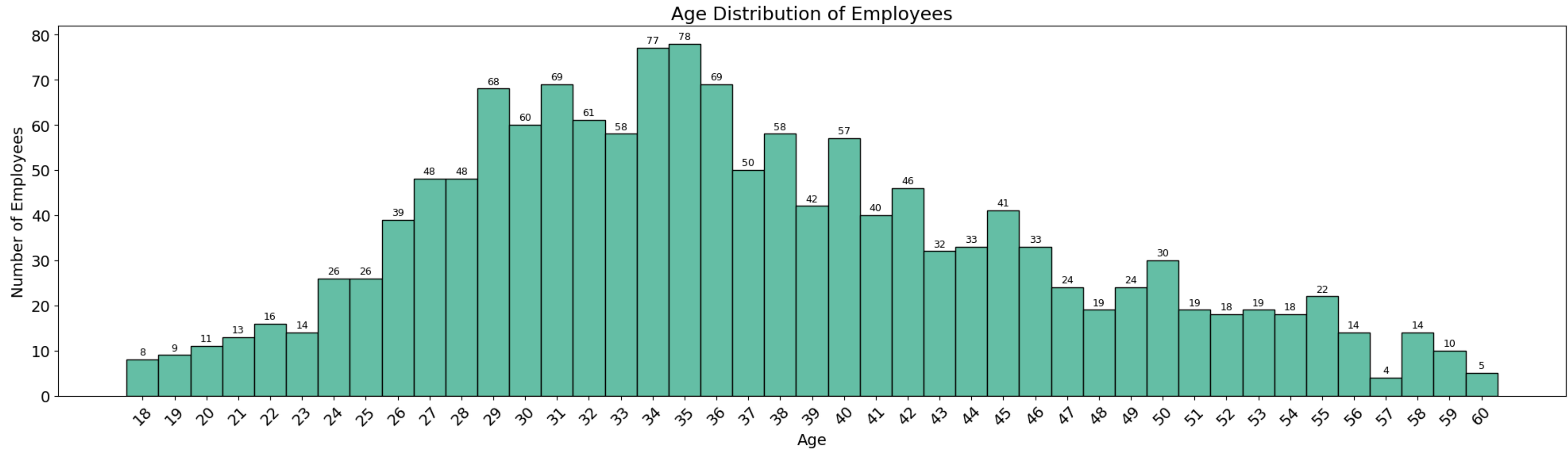
Attrition and Retention By Department



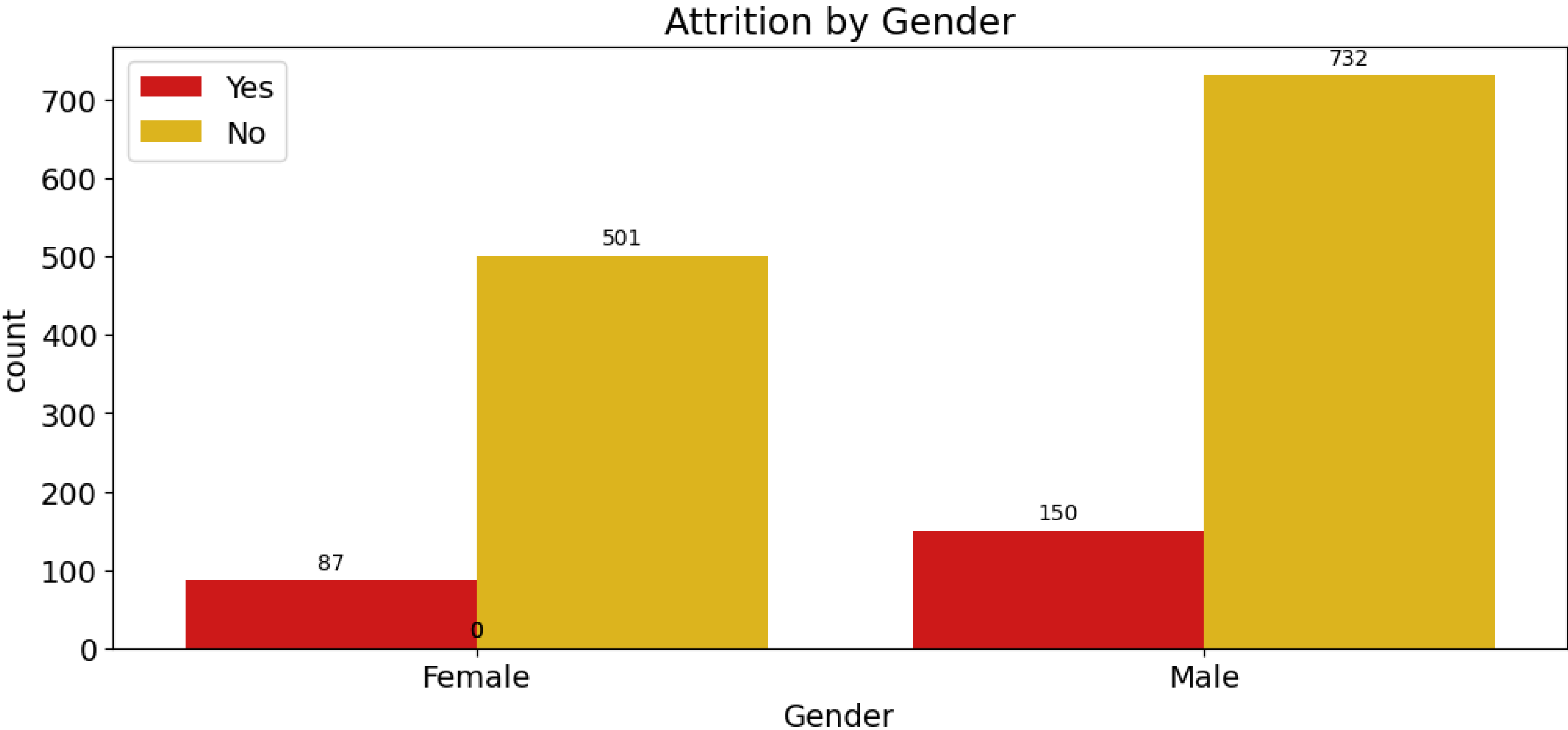


# Current Situation

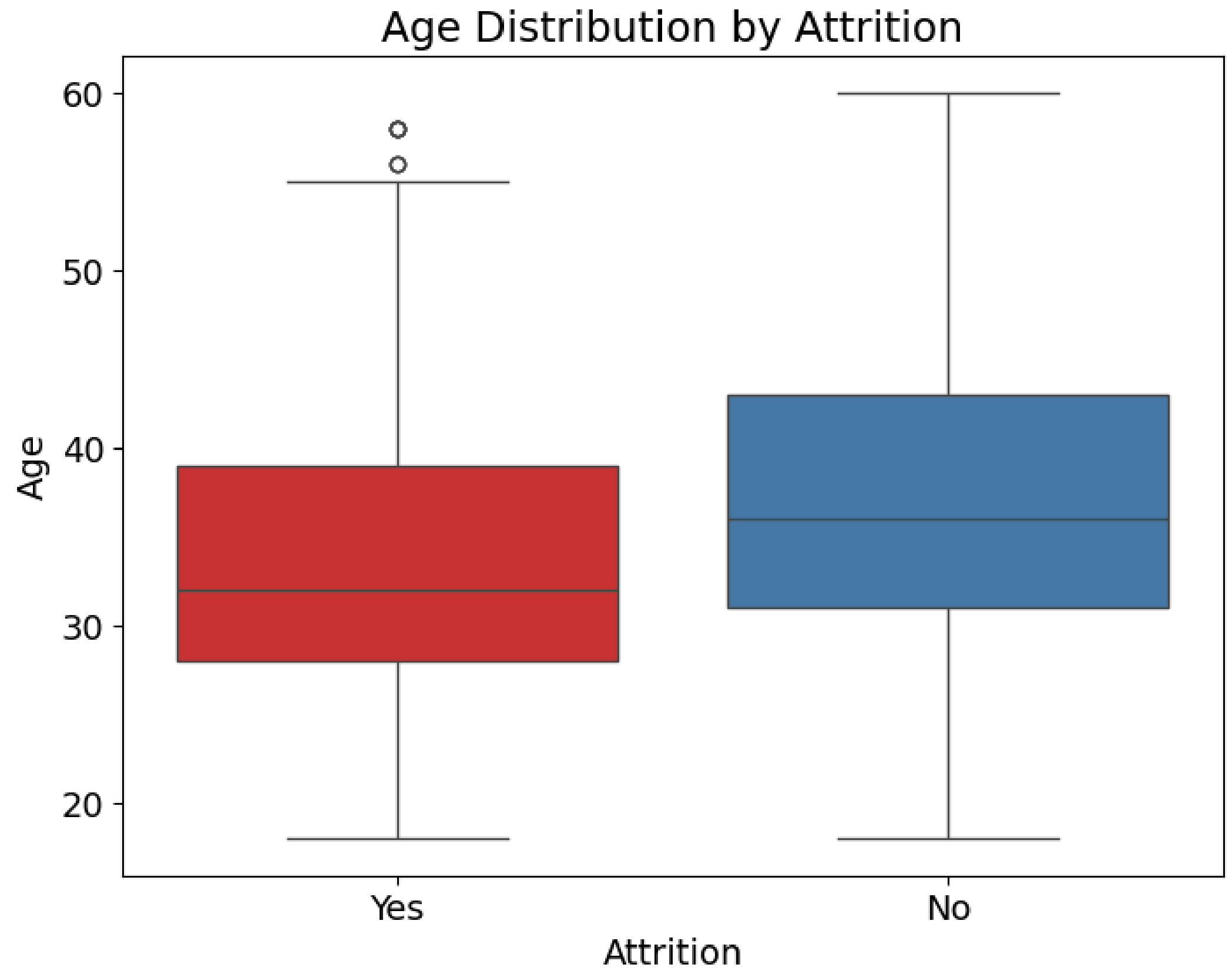
- Age Distribution



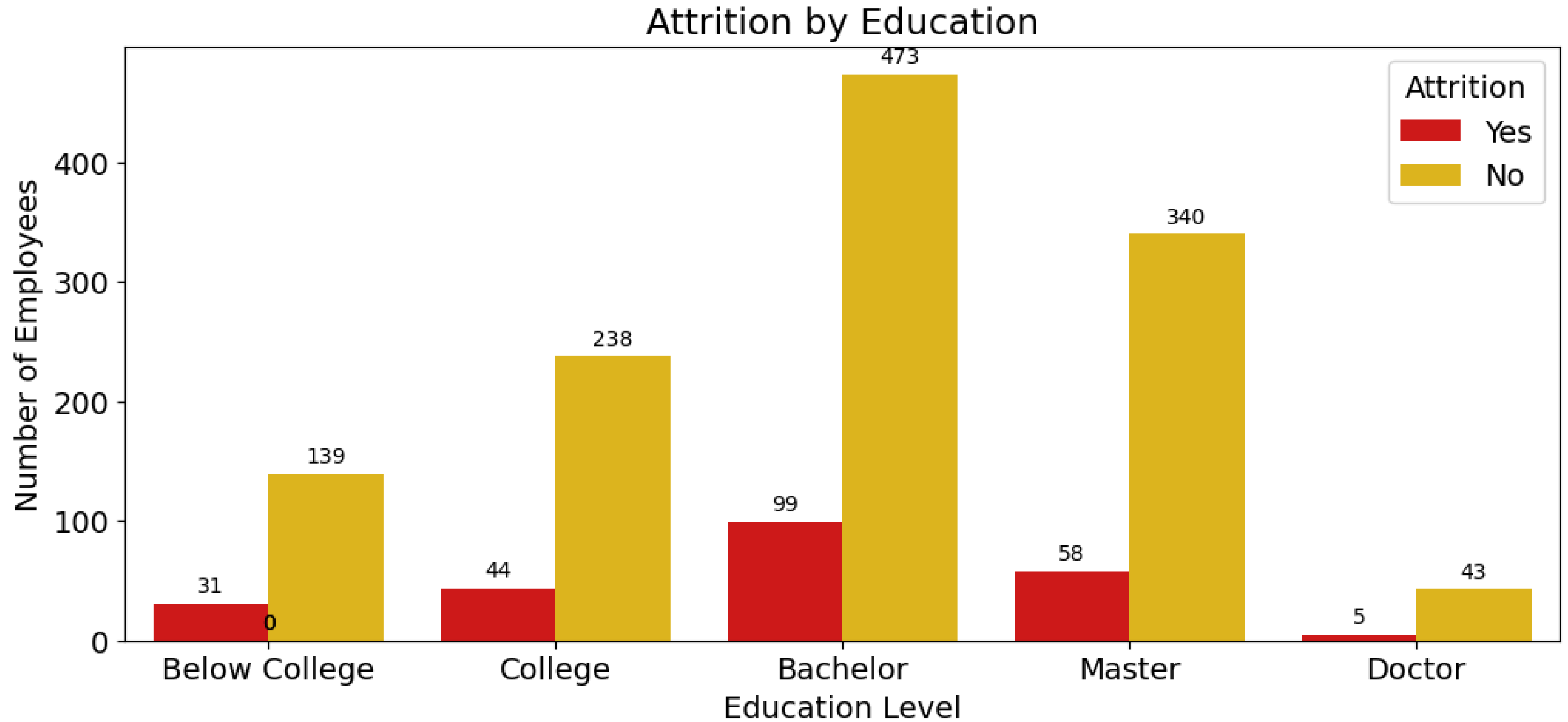
Male employees leave more  
than female employees.



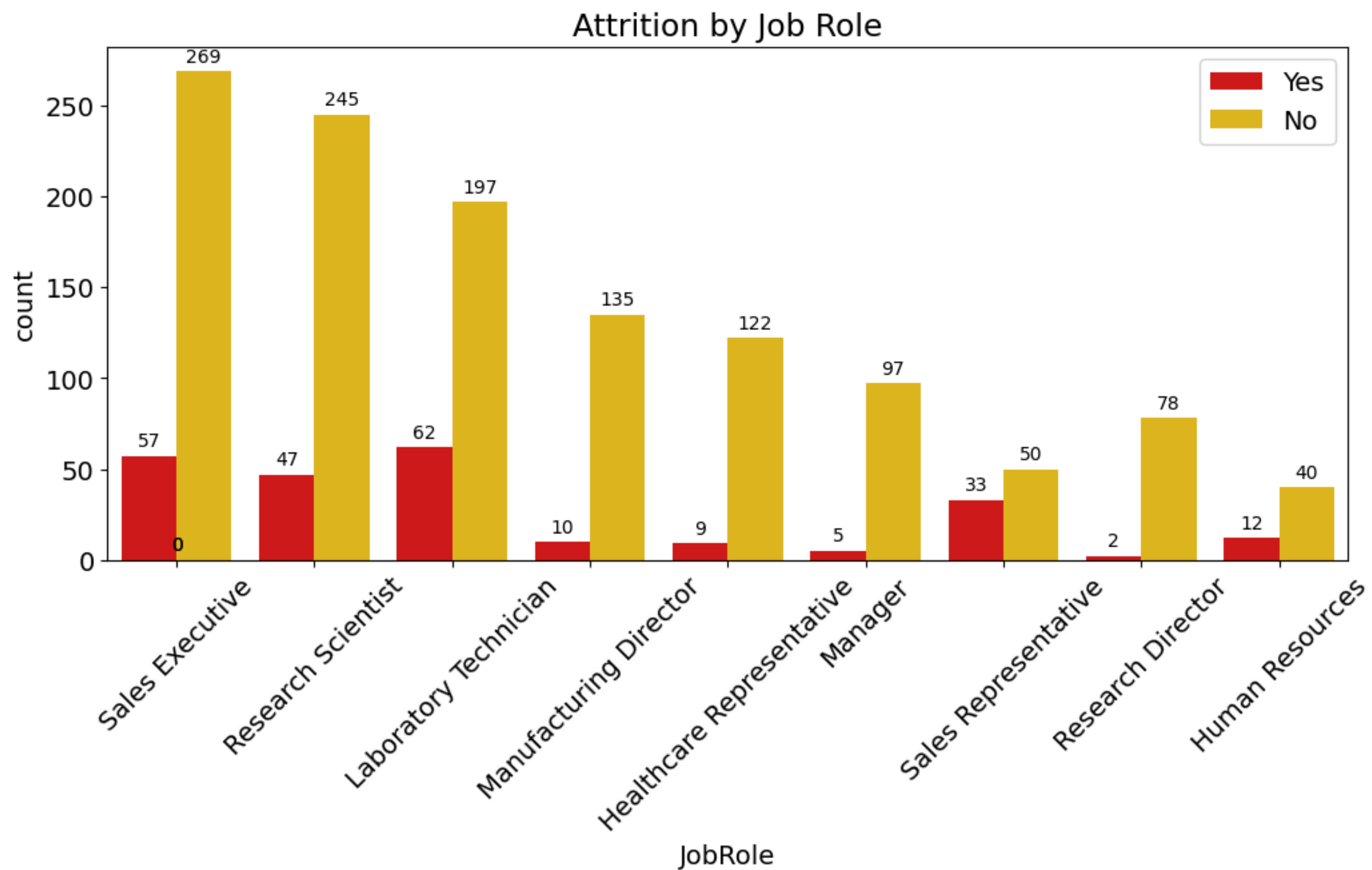
- 25-40 age group most affected



- Education: Bachelor's degree holders are at higher risk

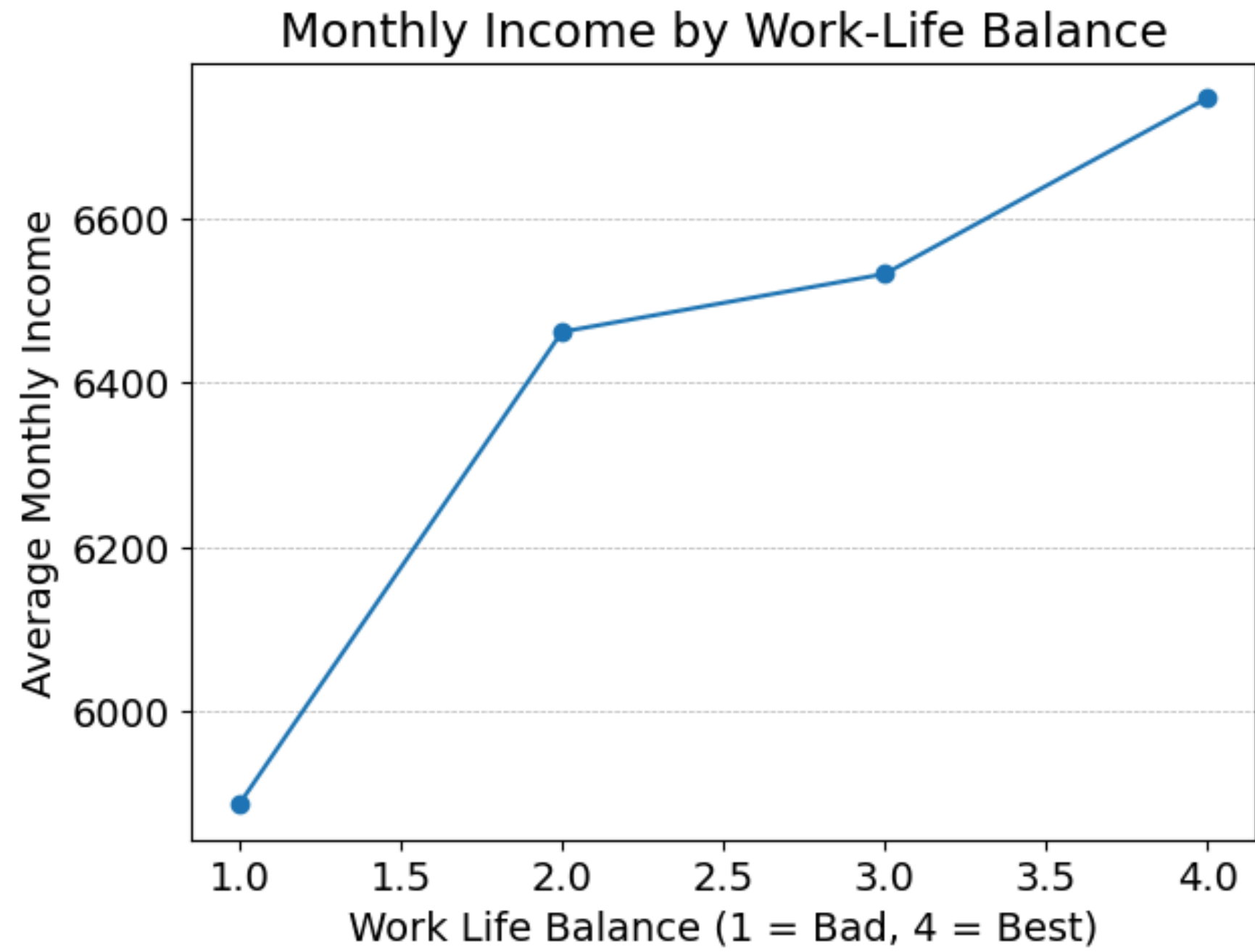
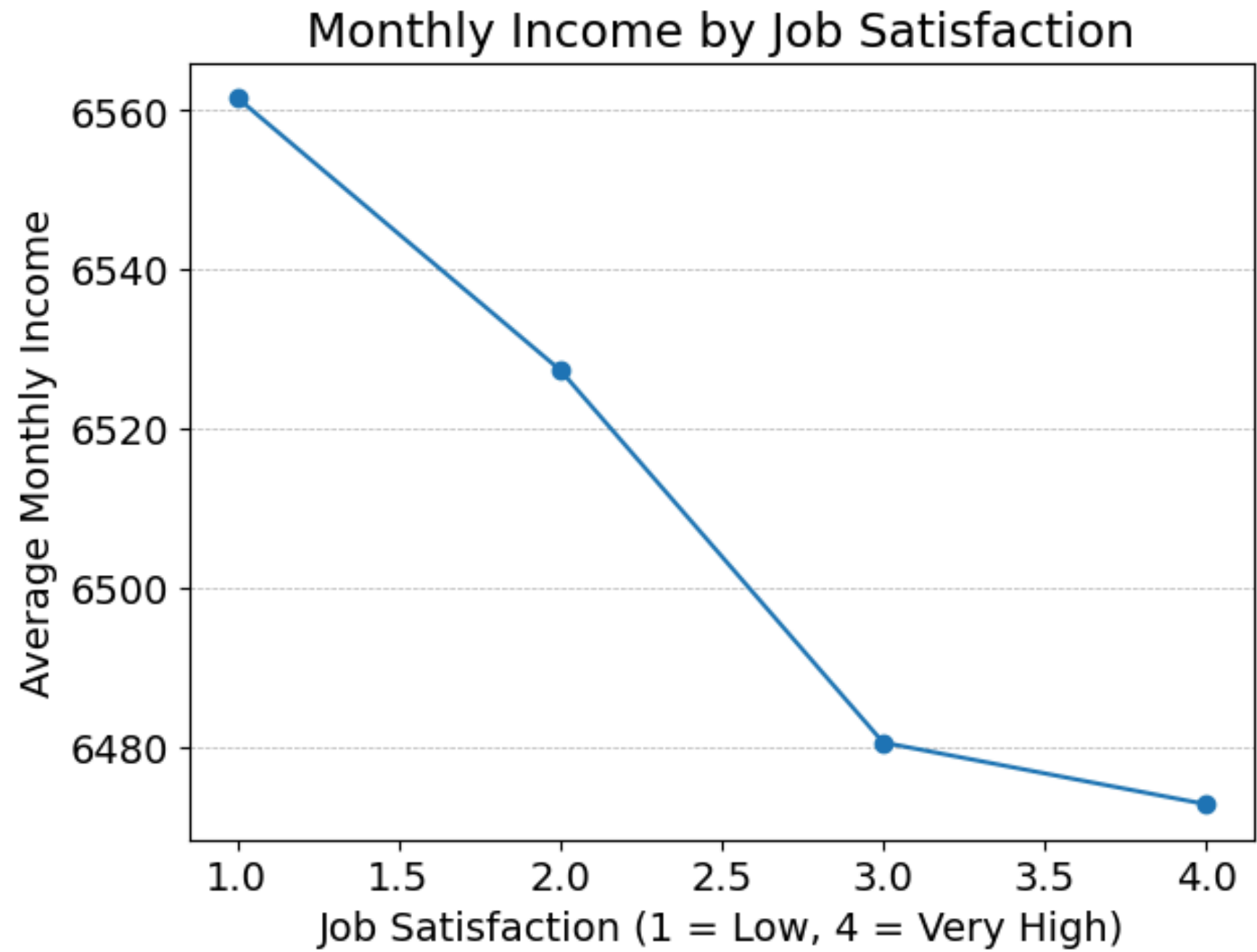


- Roles: Sales executives and representatives

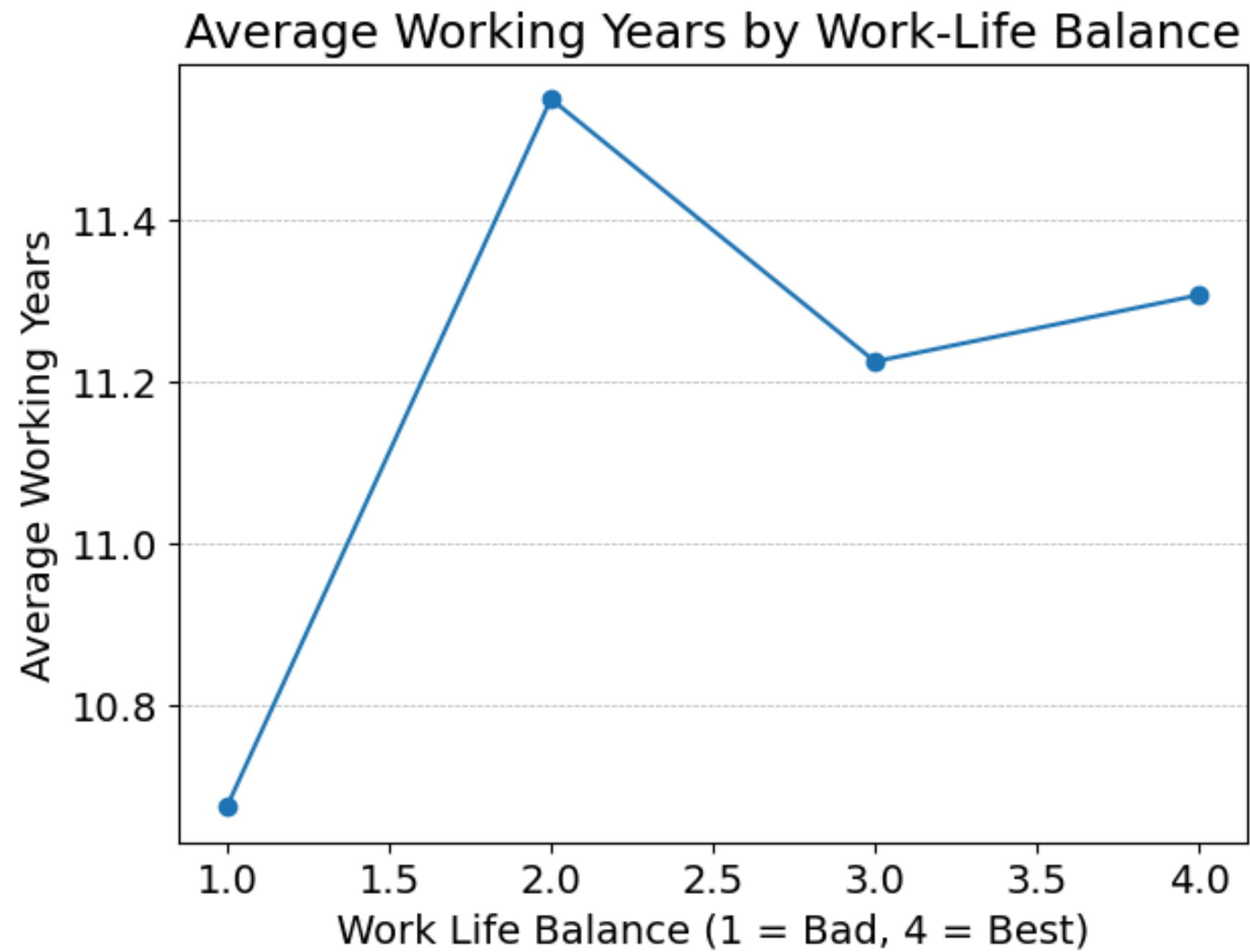
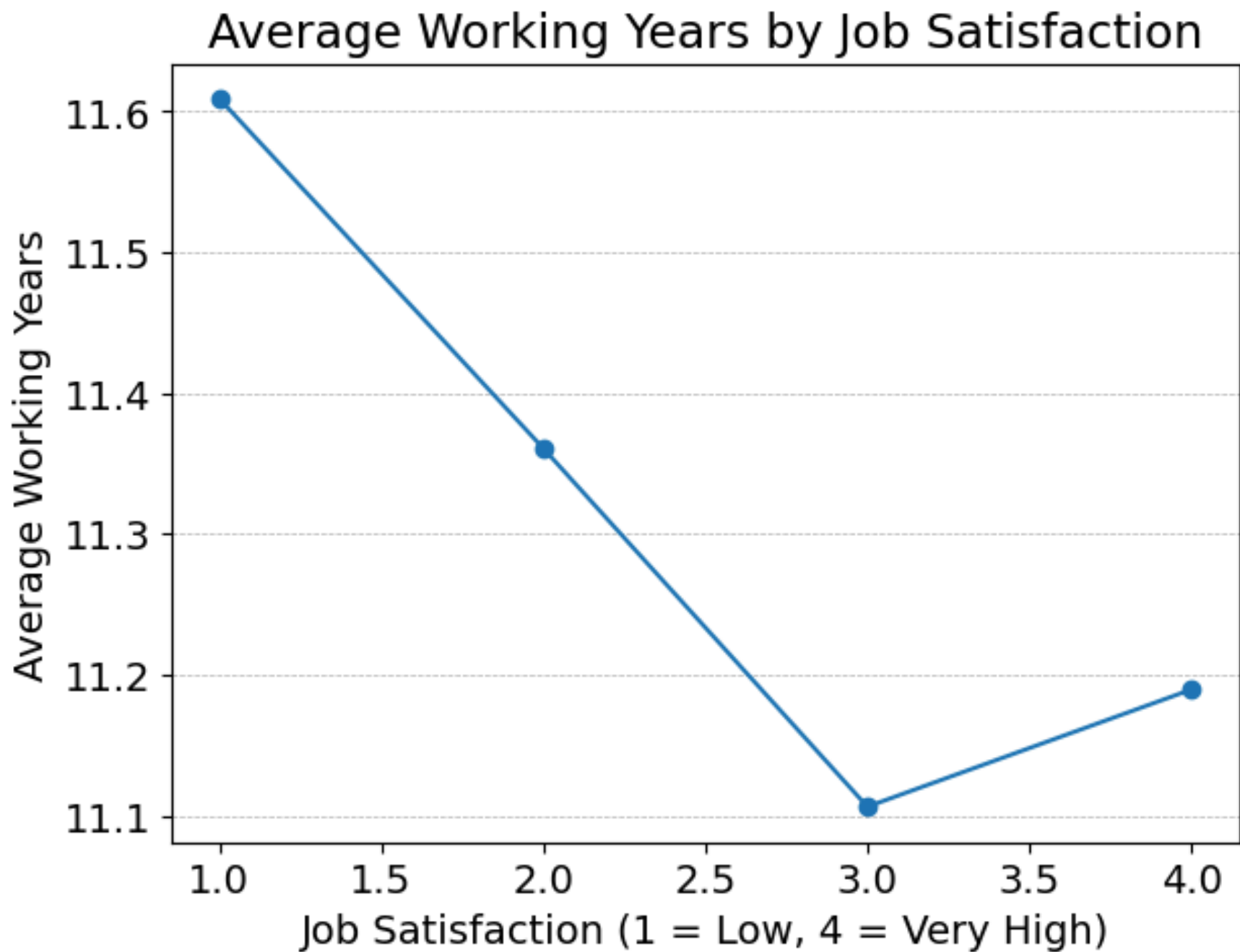




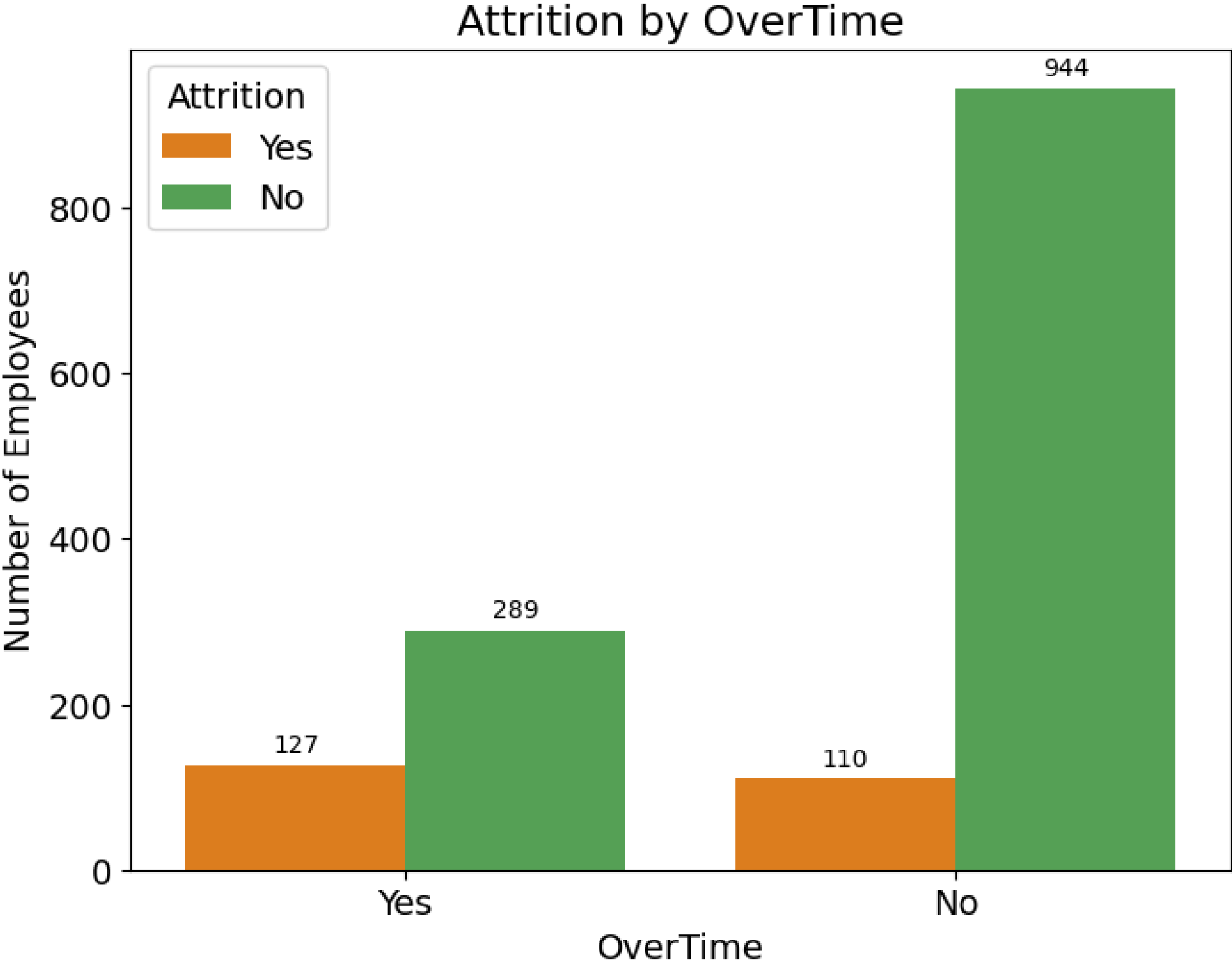
Despite having the best work-life balance,  
employees are still not satisfied with their job.



# Fewer Years, Happier Jobs; More Years, Better Balance.



The company's employees do not primarily leave due to overtime.





# Data Preparation

- Cleaned data: No missing values or duplicates

```
df.isnull().sum()
```

|                  | # 0 | ... |
|------------------|-----|-----|
| Age              |     | 0   |
| Attrition        |     | 0   |
| BusinessTravel   |     | 0   |
| DailyRate        |     | 0   |
| Department       |     | 0   |
| DistanceFromHome |     | 0   |
| Education        |     | 0   |
| EducationField   |     | 0   |
| EmployeeCount    |     | 0   |
| EmployeeNumber   |     | 0   |

| ...                     | # 0 | ... |
|-------------------------|-----|-----|
| EnvironmentSatisfaction |     | 0   |
| Gender                  |     | 0   |
| HourlyRate              |     | 0   |
| JobInvolvement          |     | 0   |
| JobLevel                |     | 0   |
| JobRole                 |     | 0   |
| JobSatisfaction         |     | 0   |
| MaritalStatus           |     | 0   |
| MonthlyIncome           |     | 0   |
| MonthlyRate             |     | 0   |

|                          | # 0 | ... |
|--------------------------|-----|-----|
| NumCompaniesWorked       |     | 0   |
| Over18                   |     | 0   |
| OverTime                 |     | 0   |
| PercentSalaryHike        |     | 0   |
| PerformanceRating        |     | 0   |
| RelationshipSatisfaction |     | 0   |
| StandardHours            |     | 0   |
| StockOptionLevel         |     | 0   |
| TotalWorkingYears        |     | 0   |
| TrainingTimesLastYear    |     | 0   |

| ...                     | # 0 | ... |
|-------------------------|-----|-----|
| WorkLifeBalance         |     | 0   |
| YearsAtCompany          |     | 0   |
| YearsInCurrentRole      |     | 0   |
| YearsSinceLastPromotion |     | 0   |
| YearsWithCurrManager    |     | 0   |

# Data Preparation

- Check for Outliers

```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns

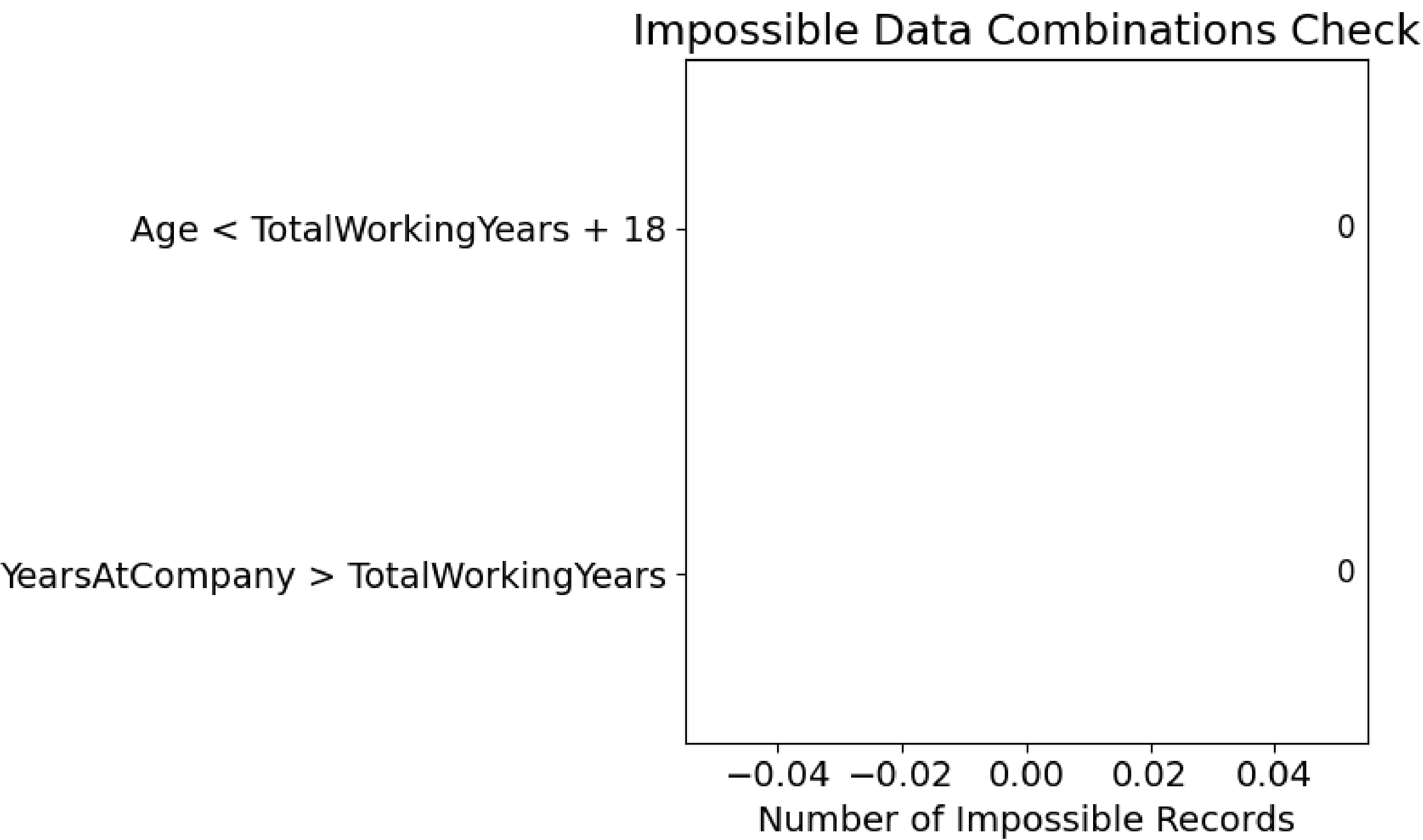
for col in num_cols:
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1
    lower_bound = q1 - 1.5 * iqr
    upper_bound = q3 + 1.5 * iqr
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

    print(f"Outliers for {col}: {len(outliers)}")
```

```
Outliers for Age: 0
Outliers for DailyRate: 0
Outliers for DistanceFromHome: 0
Outliers for Education: 0
Outliers for EmployeeCount: 0
Outliers for EmployeeNumber: 0
Outliers for EnvironmentSatisfaction: 0
Outliers for HourlyRate: 0
Outliers for JobInvolvement: 0
Outliers for JobLevel: 0
Outliers for JobSatisfaction: 0
Outliers for MonthlyIncome: 114
Outliers for MonthlyRate: 0
Outliers for NumCompaniesWorked: 52
Outliers for PercentSalaryHike: 0
Outliers for PerformanceRating: 226
Outliers for RelationshipSatisfaction: 0
Outliers for StandardHours: 0
Outliers for StockOptionLevel: 85
Outliers for TotalWorkingYears: 63
Outliers for TrainingTimesLastYear: 238
Outliers for WorkLifeBalance: 0
Outliers for YearsAtCompany: 104
Outliers for YearsInCurrentRole: 21
Outliers for YearsSinceLastPromotion: 107
Outliers for YearsWithCurrManager: 14
```

# Data Preparation

- Check for impossible combinations



# Data Preparation

- Encoded categories: Converted text to numbers for analysis

```
df['Attrition'] = df['Attrition'].replace({'No':0, 'Yes':1})
df['OverTime'] = df['OverTime'].replace({'No':0, 'Yes':1})
df['Over18'] = df['Over18'].replace({'N':0, 'Y':1})
df['Gender'] = df['Gender'].replace({'Male':0, 'Female':1})
```

```
from sklearn.preprocessing import LabelEncoder
encoding_cols=['BusinessTravel', 'Department', 'EducationField', 'JobRole', 'MaritalStatus']
label_encoders = {}
for column in encoding_cols:
    label_encoders[column] = LabelEncoder()
    df[column] = label_encoders[column].fit_transform(df[column])
```

# Data Preparation

Split data: 70% training, 30% testing

Training set size: 1029 samples

Test set size: 441 samples

Training set target distribution:

Attrition:

- No (863)
- Yes (166)

Test set target distribution:

Attrition:

- No (370)
- Yes (71)

Training attrition rate: 16.13%

Test attrition rate: 16.10%





# Data Preparation

Handled imbalance: Used SMOTE (16% vs 84% problem)

Before balancing:

Attrition:

- No (863)
- Yes (166)

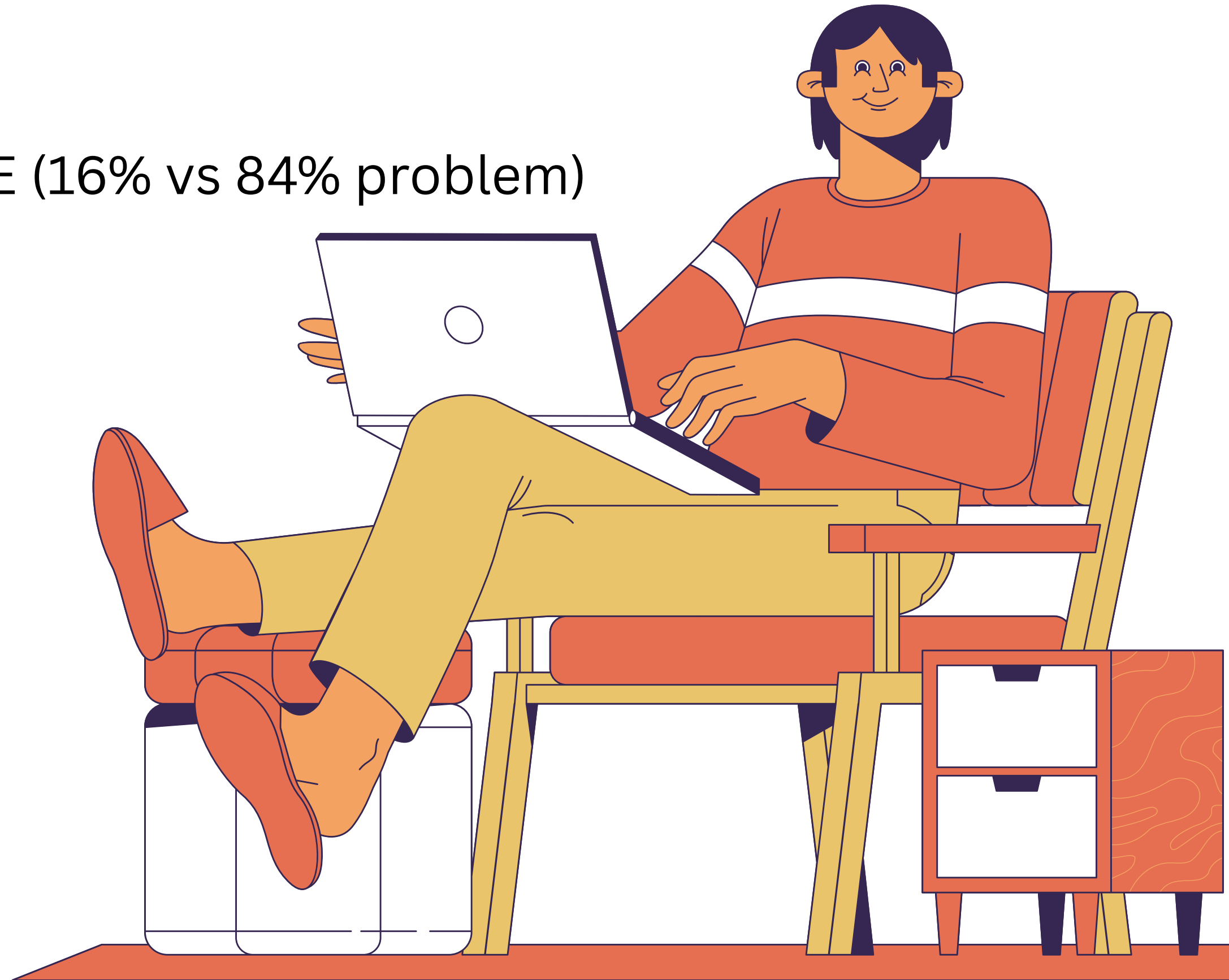
After balancing:

Attrition:

- No (863)
- Yes (863)

Attrition rate: 50.00%

Balanced training set size: 1726 samples



# Machine Learning Approach

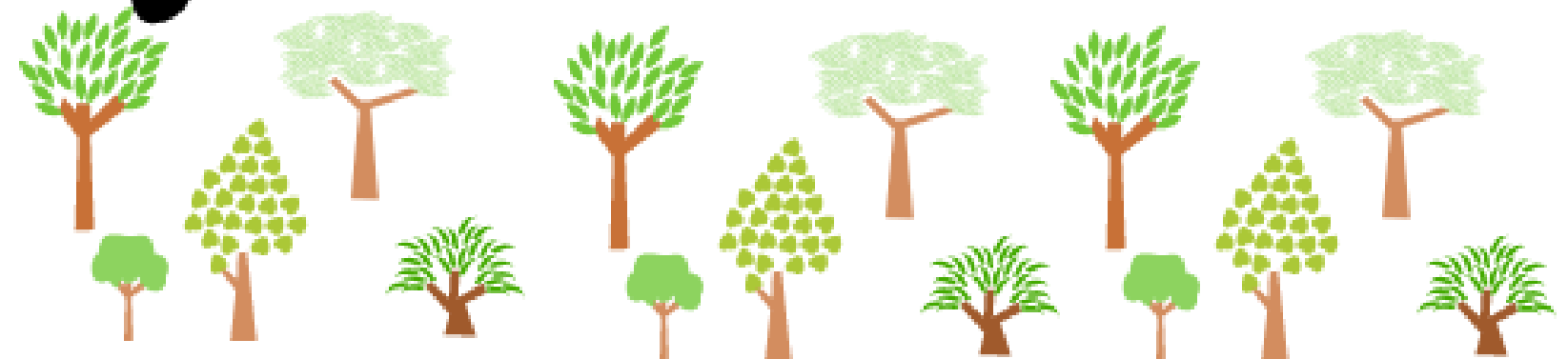
Model chosen: Random Forest w/ hyperparameter tuning

Why Random Forest?

- Handles mixed data types well
- Shows which factors are most important
- Good accuracy for business problems
- Easy to interpret results
- robust to outliers

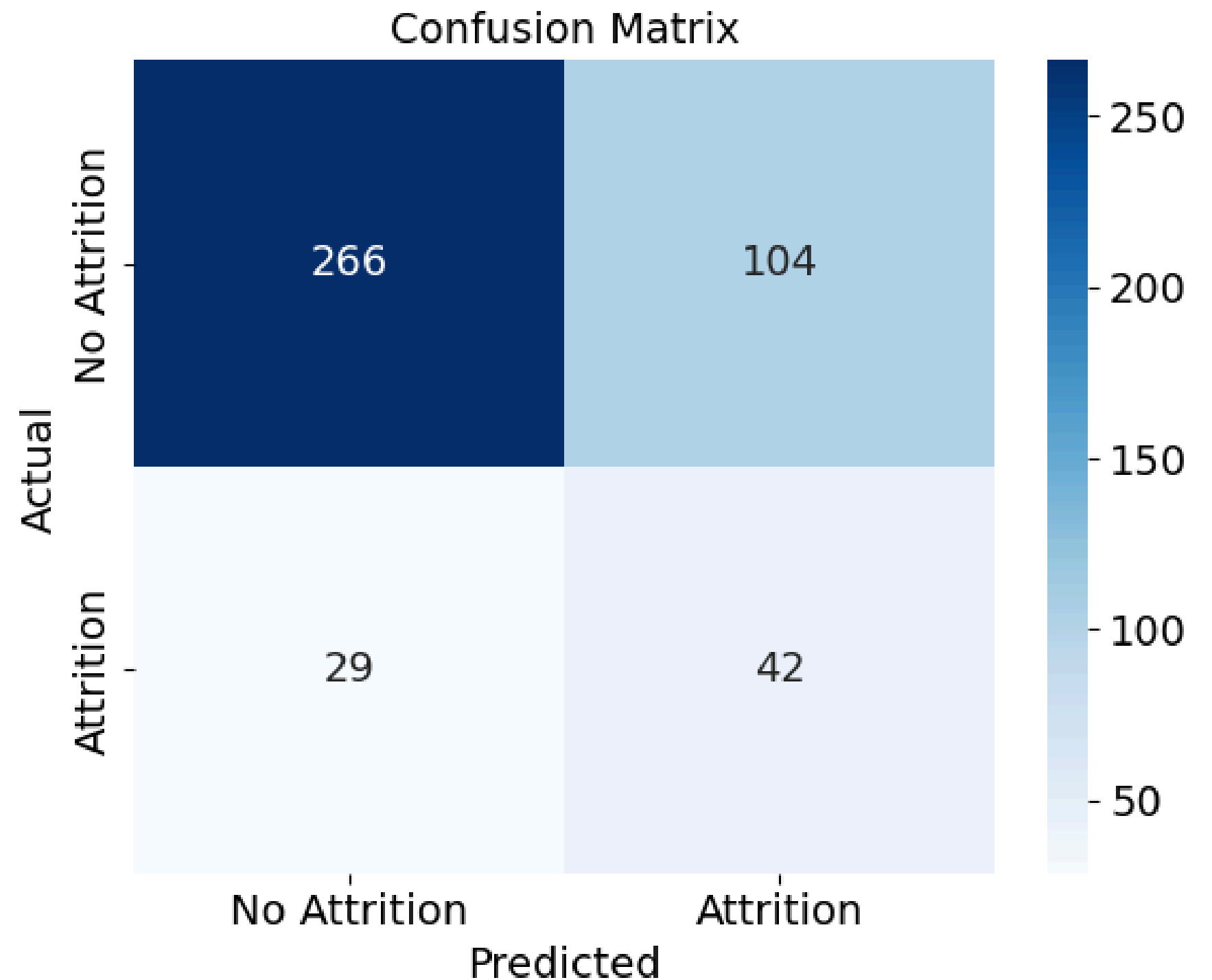


# Random Forest



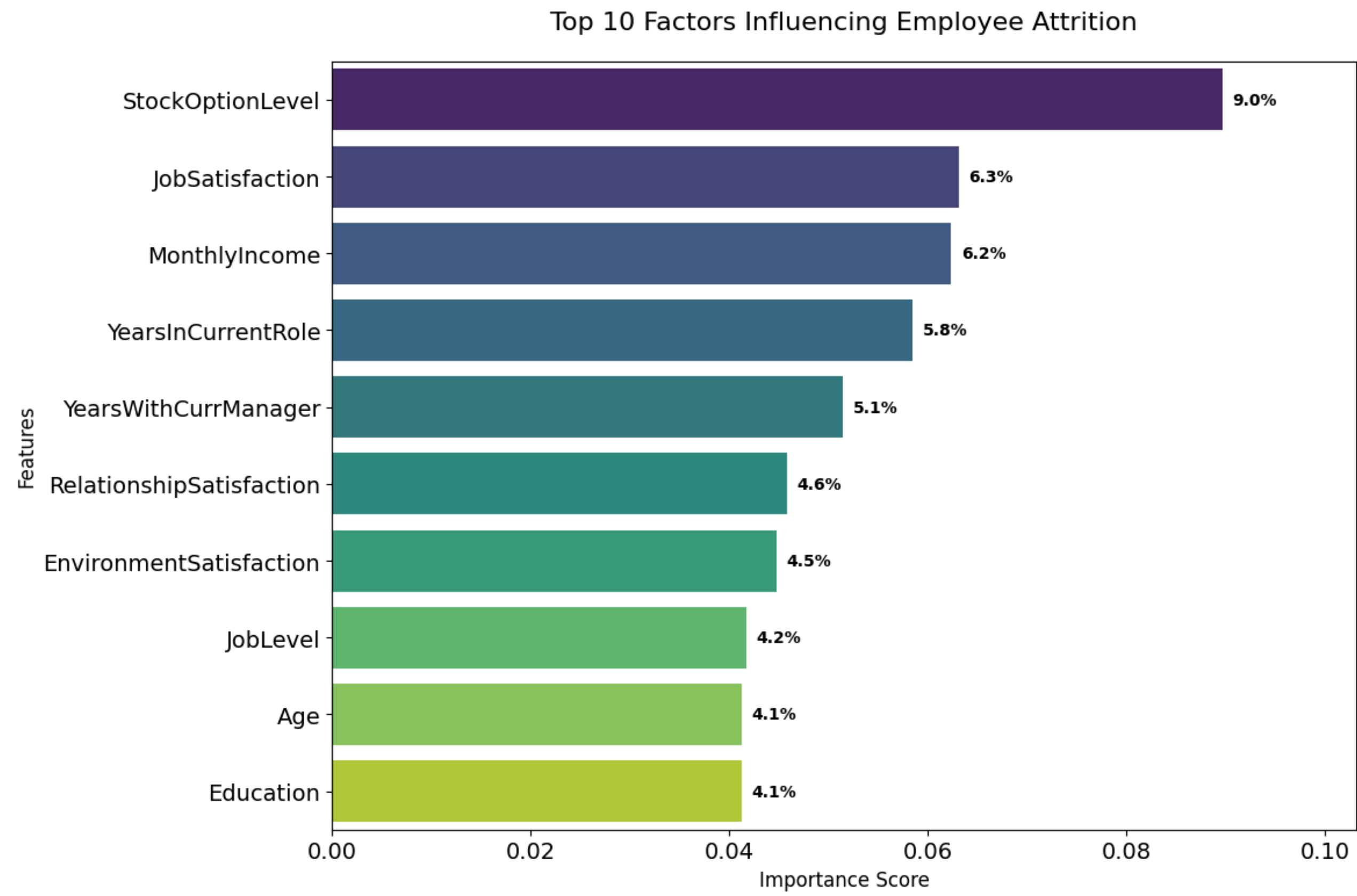
# Model Performance

- Accuracy: 0.70 or 70%
- AUC Score: 0.6979 or 69.79%
- Key strength: The Model catches 59.2% of employees who will leave

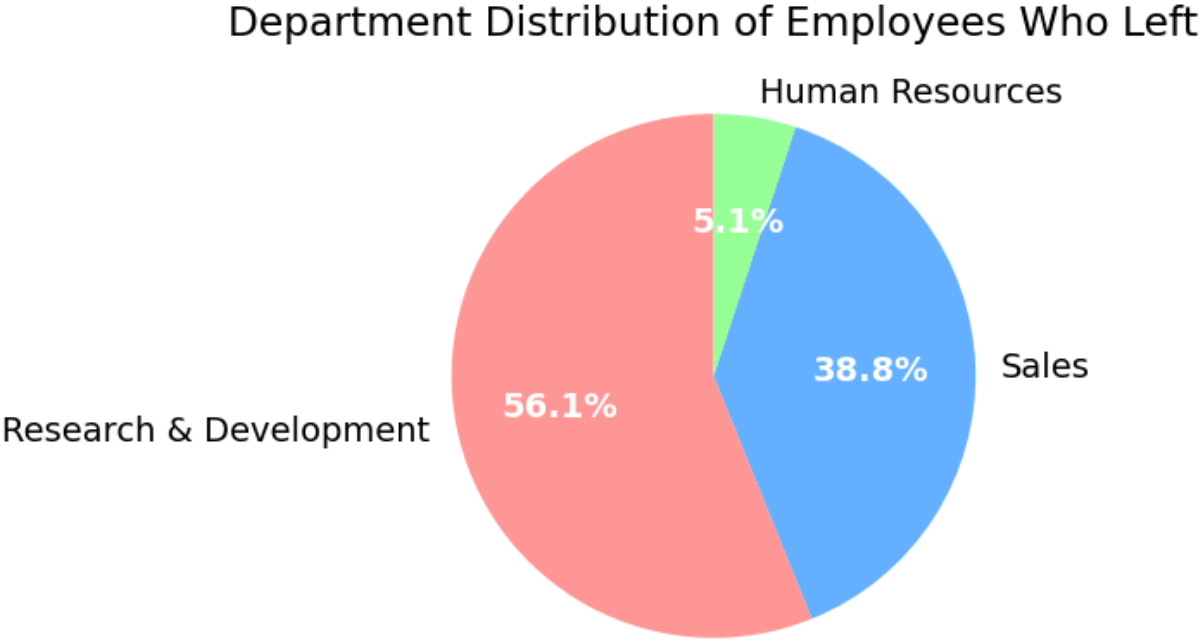
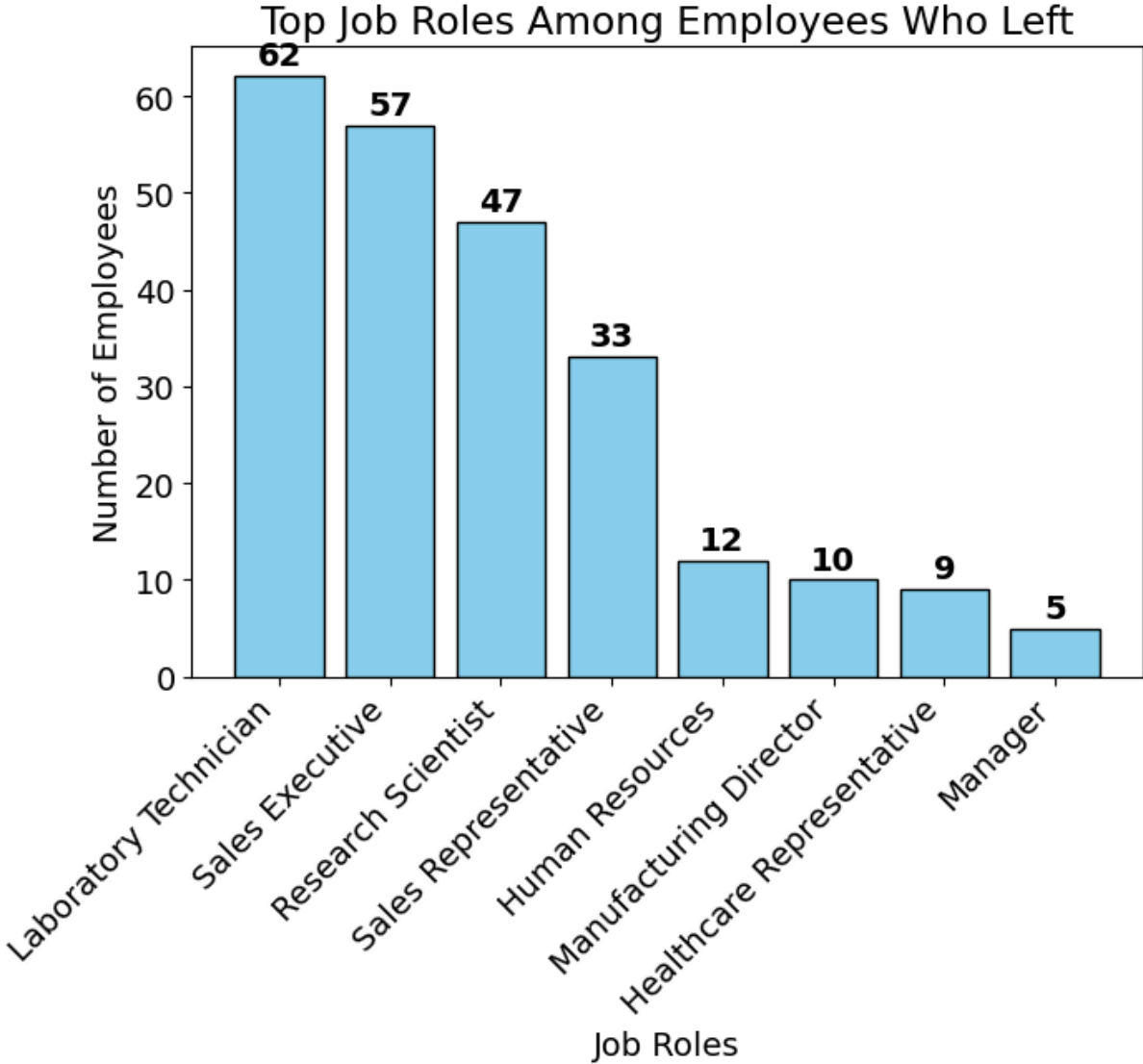
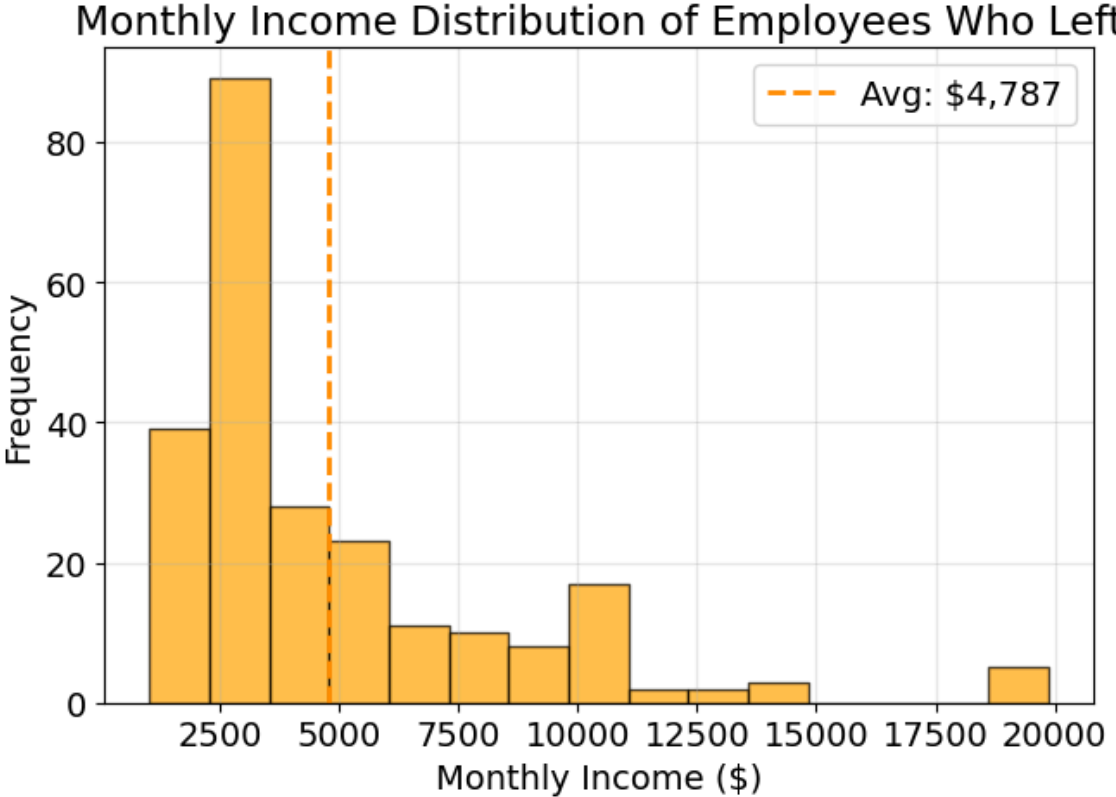
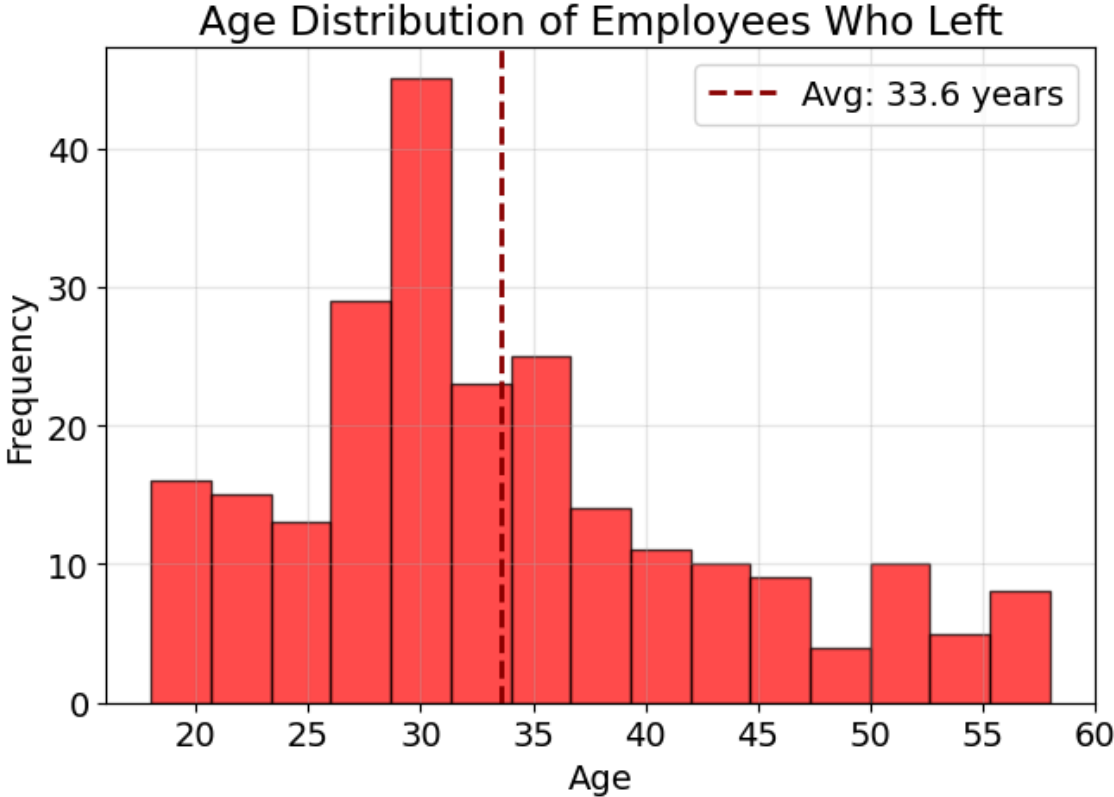




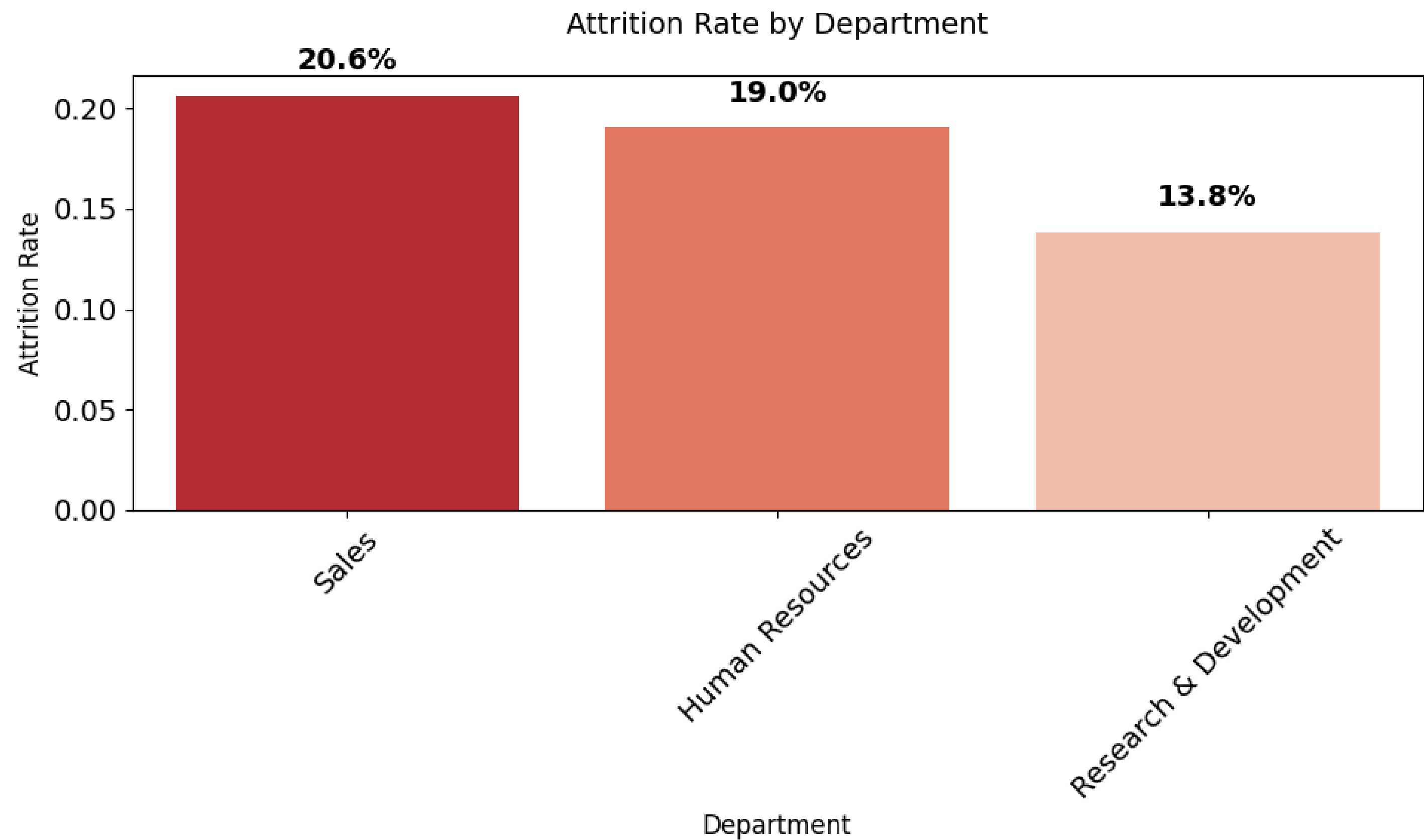
# Top Risk Factors



High-Risk Employee Profile Analysis



# Department Attrition rate Analysis



# Business Insights

**Main Driver:** StockOptionLevel (0.090)

**Department focus: Sales** needs immediate attention

**Early warning:** The Model can predict 59% of departures

## RECOMMENDATIONS:

- Focus retention efforts on the Sales department
- Address the top risk factor: StockOptionLevel
- Regular monitoring of employees with high attrition probability



# Conclusion



- Analyzed 1,470 employee records successfully
- Built a working prediction model with 70% accuracy
- Identified the top reasons for employee turnover
- Provided actionable insights for the HR team





# Thank You

